

Student Performance Analysis — Project Report

Objective

To analyze student attendance and academic marks, identify performance patterns, and generate practical insights.

Methodology

A sample dataset of 20 students was prepared. The data was analyzed using Pandas and visualized with Matplotlib. Average marks were calculated across Mathematics, Science, and English.

Results

Subject	Average Marks
Math	76.75
Science	79.30
English	78.35

The attendance-to-average-marks correlation is 1.00. This suggests a positive relationship in this sample.

Insights & Recommendations

1. Encourage regular attendance because higher attendance is associated with stronger average scores in this sample.
2. Identify students with low attendance and provide academic support early.
3. Use subject-wise performance to identify areas needing additional practice.
4. Track performance over time rather than relying on a single assessment.

Conclusion

The analysis demonstrates how basic data analytics can reveal useful patterns in student performance and support evidence-based academic recommendations.